

Tool Integrated Reasoning (TIR)

Goals

Replicate the results of the TORA paper (<https://arxiv.org/pdf/2309.17452>) in spirit, of the evaluation performance of rationale-only (i.e., CoT) vs TIR on the GSM8k and MATH datasets using the Qwen2.5 0.5B model. Create supervised training data for TIR using a subset of problems from the GSM8k and the MATH datasets.

Milestones

1. **Implement baseline.** SFT train instances of the Qwen2.5 0.5B model on the CoT data from the train splits of GSM8k and MATH datasets, and evaluate their performance on the evaluation splits of the respective datasets.
2. **Generate supervised training data for TIR.** Using the example prompt in appendix E of the TORA paper, linked above, as inspiration for formatting, create supervised training data for TIR for N=1000 examples of data from the GSM8k and MATH datasets respectively. Leverage a performant, open-source, open-weights model that is already capable of TIR to generate the data.
3. **Implement and validate TIR training pipeline.** Implement a robust TIR pipeline, inspired by the methods outlined in the TORA paper, with the exception of output space shaping, which should be skipped. Allow the user to configure training hyperparameters through command line arguments. Log training loss and accuracy per step. Assess the correctness of the training pipeline by overfitting a batch of B=8 examples.
4. **Train Qwen2.5 0.5B model on synthetic TIR data.** Run E=1 epochs of training on the synthetic TIR datasets for each of the two datasets on two separate instances of the base model.
5. **Evaluate trained models on evaluation sets of the GSM8k and MATH datasets.** Evaluate performance of trained models and compare with baseline.